

CEREBRO-X

Computational Drug-DDS Engineering Report

Field	Value
Drug Name	Rivastigmine
Molecular Class	small_molecule
Molecular Weight	250.3 Da
Recommended DDS	RVG29-PLGA-NP
Carrier Type	plga
Surface Ligand	RVG29
BBB Enhancement	100.0%
Composite Score	83.0/100
Disease Indication	CNS
Report Generated	2026-09-05 08:08 UTC
Created by	Muhammad Talaat CEREBRO-X
Modules Completed	50/62 science modules

■ This report is generated from computational models. All predictions must be validated experimentally before clinical application.

1. Drug Molecular Profile

Property	Value	Source / Method
Name	Rivastigmine	Input
MW (Da)	250.3	ChEMBL / UniProt
LogP	2.76	ADMET predictor
Half-life (days)	13.300	PubMed NLP
Protein Binding (%)	N/A	Calculated
BBB Penetration (%)	32.20	BBB model
Molecule Class	small_molecule	Input / inferred
Aqueous Solubility	N/A	ADMET
P-gp Substrate	Unknown	QSAR panel
ADMET Score	N/A	Multi-endpoint
ADMET Grade	N/A	A-F scale

1a. Drug Delivery Barriers Identified (Auto-detected)

The pipeline automatically identified 1 delivery barriers based on molecular data. Each is resolved by the recommended DDS.

[HIGH] Cardiac off-target risk (hERG/Nav/Cav channels)	
Evidence:	QSAR panel: cardiac receptor score > safety threshold
Without DDS:	Full systemic distribution -- cardiac tissues exposed
With DDS:	CNS bioavailability 53.0% with reduced cardiac exposure
DDS Solution:	CNS-targeted plga reduces systemic exposure
Scientific basis:	Brain-targeted delivery concentrates drug in CNS. Systemic free drug reduced by 3750% (encapsulation). Off-target liver

2. DDS Ranking — Composite Score Methodology

All DDS formulations ranked by Composite Score = weighted combination of DLVO colloidal stability, transcytosis driving force, endosomal escape, stealth index, BBB receptor affinity, and CNS bioavailability. Scores are physics-derived — not estimated.

DDS Name	Carrier	Ligand	Score	DLVO kT
RVG29-PLGA-NP	plga	RVG29	82.97	5.83
Tf-SLN-Stealth	solid_lipid	Transferrin	81.96	-3.49
ApoE-SLN	solid_lipid	ApoE	81.74	-0.12
RVG29-Liposome-pH	liposome	RVG29	80.98	-2.43
Tf-PEG-Liposome-Done	liposome	Transferrin	80.21	-5.69
Lf-Micelle-Thermo	micelle	Lactoferrin	75.09	-5.05
Bare-PLGA	plga	None	74.70	23.22
PEG-Polymer-Micelle	micelle	None	69.78	-5.07

2a. Recommended DDS — Detailed Profile

Parameter	Value	Biophysical Meaning
Composite Score	83.0	Overall Drug+DDS suitability (0-100)
BBB Enhancement	100.0%	% of dose crossing BBB via receptor transcytosis
CNS Bioavailability	53.0%	% dose reaching brain as free drug
DLVO Stability	5.8 kT	>25kT = colloidally stable in blood
Endosomal Escape	0.64	Fraction escaping lysosomes (0-1)
Stealth Index	1.00	MPS evasion (0=opsonised, 1=invisible)
Protein Corona	N/A nm	Corona thickness; thicker = faster clearance
MPS Clearance	N/A h	Hours before liver/spleen removes DDS
Payload Efficiency	N/A%	% dose delivered as active drug to CNS
CARPA Risk	0.42	Complement activation risk (0=safe)

3. PBPK-CNS Digital Twin — 6-Compartment Simulation

Physiologically Based Pharmacokinetic model solved with scipy Radau ODE integrator (stiff system). Compartments: Plasma, BBB endothelium, Brain ISF, Brain cells, CSF, Peripheral. All parameters from molecular data — not estimated.

PK Metric	Value	Compartment	Source
Model	6-compartment ODE (Radau stiff solver)	—	CEREBRO-X PBPK
Cmax (brain)	0.214000 µg/mL	Brain/CNS	Peak brain concentration
AUC (brain)	7.14060 µg·h/mL	Brain/CNS	Total brain exposure
AUC (plasma)	6.920 µg·h/mL	Plasma	Systemic exposure
Kp,brain	1.03180	Brain/Plasma	Partition coefficient
BBB penetration	0.0000%	BBB	
Kp,uu brain	2.20102	Unbound	Unbound Kp (pharmacodynamically relevant)
t_above 10%Cmax	72.0 h	Brain ISF	Duration above therapeutic threshold
Glymphatic t1/2	25.0 h	Brain	Clearance half-life in brain
BBB PS_in	778.982 mL/h	BBB	Permeability-surface area product
BBB PS_out	292.118 mL/h	BBB	Efflux transport rate
Disease state	healthy	BBB	BBB integrity modifier applied

3a. PBPK Time-Course (sampled points)

t (h)	Plasma (µg/mL)	Brain ISF (µg/mL)	Ratio (Kp)
0.0	0.2500	0.00000	0.00000
6.5	0.1888	0.19464	1.03101
13.0	0.1543	0.16100	1.04365
19.5	0.1270	0.13394	1.05470
26.0	0.1052	0.11200	1.06429
32.5	0.0877	0.09406	1.07250
39.3	0.0730	0.07877	1.07971
45.8	0.0613	0.06660	1.08552
52.3	0.0518	0.05644	1.09037
58.8	0.0438	0.04793	1.09439
65.3	0.0371	0.04077	1.09770
72.0	0.0314	0.03453	1.10051

4. In-Silico Drug Release Profile

Release model: First Order | Release order: pH-triggered (endosomal) | Max EE: 75%

Metric	Blood (pH 7.4)	Endosomal (pH 5.5)	Significance
t50 (50% release)	23.2 h	36.2 h	Faster endo = better escape
t90 (90% release)	48.0 h	N/A	Complete release window
Rate constant k	0.0300 /h	0.0192 /h	Higher endo = pH-responsive
Max released	75%	75%	= encapsulation efficiency

5. Shelf-Life & Degradation Predictor

Grade: MARGINAL (3-6 months) | t90 = 122 days | Dominant pathway: Drug leakage | Storage: 4 degC

Pathway	Rate (k, /day)	Relative %
Hydrolysis	0.000061	7%
Oxidation	0.000160	19%
Aggregation	0.000273	32%
Drug leakage	0.000368	43%

6. Colloidal Stability — DLVO Theory + Protein Corona

DLVO potential energy $V_{\text{total}} = 5.8 \text{ kT}$ (UNSTABLE ~~x~~ — REFORMULATE) | Threshold: $>25 \text{ kT}$ prevents aggregation in blood.

Parameter	Value	Equation	Significance
DLVO V_{total} (kT)	5.8	$V_{\text{vdW}} + V_{\text{EDL}}$	$>25\text{kT} = \text{stable}$
V_{vdW} (attraction)	N/A	$-A \cdot R / (12h)$	Hamaker constant
V_{EDL} (repulsion)	N/A	$64\pi \epsilon R \gamma^2 e^{-\kappa h}$	Zeta potential
Debye length (nm)	N/A	$1/\kappa$	Ionic screening
Protein corona (nm)	N/A	Langmuir model	Stealth reduction
Opsonin index	N/A	IgG + C3b binding	$<0.3 = \text{good}$
Stealth index	1.000	PEG brush model	$0.6+ = \text{good}$
MPS clearance (h)	N/A	Michaelis-Menten	$>12\text{h}$ adequate

7. Nanotoxicity & Immunogenicity Screening

Immunogenicity grade: **CAUTION** | Score: 47/100

Test	Score / Result	Risk	Threshold
CARPA (complement)	0.945	HIGH	$>0.6 = \text{HIGH}$
Anti-PEG antibody	0.250	LOW	$0.5+ = \text{caution}$
MPS uptake	0.307	—	$>0.6 = \text{rapid clearance}$
Cytokine storm	—	LOW	Charge-dependent
Platelet activation	—	LOW	Cationic $>200\text{nm}$

Mitigations:

→ Consider pre-medication with antihistamines (H1/H2 blockers)

8. Off-Target Toxicity — 50-Receptor QSAR Panel

Overall: CRITICAL | Cardiac risk: YES ■ | Hepatic risk: YES ■ | CNS off-target: YES ■

Receptor	Free Drug Score	In-DDS Score	Risk
hERG_K+	0.228	0.103	LOW
Nav1.5_Na+	0.526	0.237	HIGH
Cav1.2_Ca2+	0.177	0.080	LOW
beta1_AR	0.301	0.136	MODERATE
CYP3A4_inhib	0.160	0.072	LOW
CYP2D6_inhib	0.667	0.300	HIGH
CYP2C9_inhib	0.210	0.095	LOW
CYP1A2_inhib	0.130	0.059	LOW
CYP2C19_inhib	0.040	0.018	LOW
DAT_dopamine	0.456	0.205	MODERATE
SERT_serotonin	0.419	0.189	MODERATE
NET_norepineph	0.405	0.182	MODERATE
MAO-A	0.006	0.003	LOW
MAO-B	0.330	0.149	MODERATE
D2R_dopamine	0.663	0.298	HIGH
5HT2A_serotonin	0.532	0.239	HIGH
GABA-A	0.240	0.108	LOW
NMDA_receptor	1.000	0.450	HIGH
nAChR_alpha4	0.320	0.144	MODERATE
H1_histamine	0.520	0.234	HIGH

High-risk flags (10):

- Nav1.5_Na+: HIGH (free_drug=0.53)
- CYP2D6_inhib: HIGH (free_drug=0.67)
- D2R_dopamine: HIGH (free_drug=0.66)
- 5HT2A_serotonin: HIGH (free_drug=0.53)
- NMDA_receptor: HIGH (free_drug=1.00)
- H1_histamine: HIGH (free_drug=0.52)
- ERalpha_estrogen: HIGH (free_drug=0.74)
- ARalpha_androgen: HIGH (free_drug=0.50)

9. Advanced Science Modules — Key Outputs

9a. Competitive DDS Landscape (ClinicalTrials.gov)

Our DDS: BBB Score = 100 | Position: SUPERIOR | Active trials found: 4

NCT ID	Title	Phase	N
NCT04899908	Stereotactic Brain-directed Radiation With or	['PHASE2']	134
NCT06855368	Biomarkers for Monitoring Dementia Progressio	['?']	300
NCT06846658	Exploring the Olfactory Mucosa, Blood and Uri	['?']	180
NCT07667387	Study of LNP.UCD.ABE in Patients With Urea Cy	['PHASE1', 'PHASE2']	7

9b. Patient Subgroup Stratification

Overall response: 85.9% | Best: Very elderly (>80) (88%) | Worst: Young adults (18-40) (84%) | Recommendation: Highest efficacy in Very elderly (>80) (predicted response 88%). Dose reduce by

Subgroup	Pop. Freq.	CNS Bioavail	Response%	Tox Risk	Dose Adj.
Young adults (18-40)	20% of patients	44.2%	83.5%	LOW	x1.29 standard dose
Middle-aged (40-65)	35% of patients	50.3%	86.0%	LOW	x1.14 standard dose
Elderly (65-80)	30% of patients	51.5%	86.4%	LOW	x0.72 standard dose
Very elderly (>80)	15% of patients	55.6%	87.7%	MODERATE	x0.46 standard dose

9c. Quantum Coherence Transport Model

WKB tunneling probability = 4.108e-10 | Barrier = 1.62 kcal/mol | Classical prob = 0.07219 | Quantum tunneling probability = 4.11e-10 (negligible contribution). Classical ba

9d. Lysosomal Trafficking Predictor

Cytosolic route: 42% | Lysosomal degradation: 56% | Nuclear entry: 2% | Concern: HIGH -- significant lysosomal degradation expected | Mitigation: Add proton sponge polymer (PEI or PAMAM) to improve endosomal escape

10. Synthetic Clinical Trial — N=500 Virtual Patients

Monte Carlo simulation of Phase 1 trial. Patient covariates: age, weight, CYP3A4 genotype, renal function, BBB integrity by age. Pharmacokinetics computed per-patient from PBPK model.

NO-GO / REFORMULATE Go/No-Go	100.0% Response Rate	91.2% Severe AE	0.07 mg/kg Optimal Dose	500 N Patients
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Endpoint	Result	Benchmark	Status
Overall response rate	100.0%	>60% = GO	✓
Severe AE rate	91.2%	<5% = GO	✗
Young adults (<65)	100.0%	—	—
Elderly (>65)	100.0%	—	—
Mild AE	8.8%	—	—
Renal AE	26.8%	<15%	✗

Recommendation: Predicted Phase 1 success rate: 100%. Dose 0.07 mg/kg recommended. Monitor renal function in 27% of elderly patients.

11. Manufacturing & Sterilization Modules

11a. Terminal Sterilization Survivability

Recommended method: Autoclave 121C | Cost implication: Standard (autoclave preferred) | Feasible methods: autoclave 121C, gamma 25kGy, aseptic filtration 022um, VHP H2O2

Method	Survives?	Detail
autoclave 121C	YES ✓	Thermally stable
gamma 25kGy	YES ✓	Survives gamma sterilization
aseptic filtration 022um	YES ✓	Size 130 nm < 220 nm -- filterable
VHP H2O2	YES ✓	PEG provides surface protection

11b. Lyophilization Cycle Optimizer

Parameter	Value	Standard
Tg' (cryo.	-30°C	Formulation-specific
Primary drying T	-32.0°C	Must be < Tg'
Primary P	0.1478 mbar	0.5×P_ice
Primary time	25.6 h	Sublimation model
Secondary T	25.0°C	+25°C
Cycle total	41.6 h	Include ramp time
Cake collapse	OK: primary drying fixed at Tg'-2degC safety margin (-32 degC)	Must say OK

11c. Cryo-Chain Excursion Predictor

Excursion: -20.0°C for 4.0h | EE: 75.0% → 74.2% (loss: 0.8%) | Decision: RELEASE BATCH | Confidence: 98%

11d. Adversarial Stress-Testing

Grade: 4/5 scenarios passed -- CONDITIONAL | 4/5 scenarios passed.

Scenario	Pass/Fail	Detail
Fever 40C	FAIL ✗	Tm=42.0degC -- RISK: may partially melt at fever temperature
Shear Stress	PASS ✓	Carrier withstands capillary shear
Oxidative Stress	PASS ✓	Polymeric carrier -- relatively resistant to oxidative stress
High Protein Plasma	PASS ✓	Stealth index 1.00: adequate MPS evasion
Repeated Dosing	PASS ✓	Anti-PEG IgM may reduce t1/2 to 70% of first dose after re-dosing. Mon

12. Regulatory Compliance & Supply Chain

12a. FDA 21 CFR Part 11 Compliance

Status: NOT VERIFIED -- no audit trail entries found for this trial. log_computation() must be called during computation to build a compliant audit trail; this trial has none. | Audit entries: 0 | Data integrity: N/A -- no audit entries to hash-chain

12b. Geopolitical Supply-Chain Risk Assessment

Overall supply risk: HIGH | Supply chain score: 0/100 | Recommendation: Establish 6-month buffer stock for high-risk materials

Material	Risk	Source	HHI Index	Mitigation
RVG29 peptide	VERY HIGH	Custom synthesis (1-2 suppliers)	0.85	Qualify alternative supplier

12c. Digital Pharmacovigilance — Post-Elimination Fate

Unchanged excretion: 6.2% | Eco-persistence: 31.7 days | Environmental risk: HIGH | Renal dose adjustment: Reduce by 50%

13. Literature Citations — PubMed Auto-Mined

Top relevant publications retrieved live from PubMed E-utilities API based on drug name + carrier type. These papers confirm or contextualize the computational predictions above.

[1]

[2]

14. Biophysical Binding & CNS-Specific Modules

14a. FEP+ Binding Affinity (LIE approximation)

Ligand: rvg29 | ΔG_{single} = -12.50 kcal/mol | $\Delta G_{\text{avidity}}$ = -10.09 kcal/mol | K_d = 78 nM (Moderate (10-100nM)) | n_{ligands} = 42474 | Residence time = 12882219.1 s

14b. Glymphatic Clearance Simulation

ECM binding index = 0.740 | $t_{1/2}$ waking = 12.4 h | t_{90} clearance = 41.4 h | GOOD: particle size > ECM pore -- extended CNS retention

14c. Microglial Activation & Neuroinflammation

Neuroinflammation score = 0.043 | Risk: LOW -- safe for CNS application | IL-6 fold-change = 1.4× | $\text{TNF-}\alpha$ = 1.3×
→ No mitigation needed

14d. FUS-Responsive Nanocarrier Design

Frequency = 0.5 MHz | MI target = 0.40 | BBB open window = 42 min | FUS uptake = 48.0% | Inject 130 nm DDS 5 min before FUS. FUS opens BBB for 42 min. Predicted 48% enha

14e. 62-Principle C+ Flow — Class A Surrogate (Top-1 DDS)

Top-1 DDS: RVG29-PLGA-NP **Composite:** 83.0/100 **Verdict:** EXCELLENT

RVG29-PLGA-NP: composite CNS-principle score = 83.0/100 (EXCELLENT). Carrier: plga, size 0.0 nm, ζ +0.0 mV, ligand: RVG29. Strongest group: G4 Safety (91.6/100); weakest: G8 Translational (0.0/100). Evaluated against all 57 Class A surrogate principles.

CNS Principle Group Rollups

CNS Group	Score (0-100)
G1 CNS Delivery	87.8
G2 Release Kinetics	58.4
G3 Stability	71.1
G4 Safety	91.6
G5 Glymphatic BBB	61.7
G6 Manufacturability	81.0
G7 DrugDDS Fit	65.8
G8 Translational	0.0

Top Strengths (Class A Surrogate)

Principle	Score	Method	Reference
P02	100.0	BW ^{0.75} scaling + class-specific adjustment	
P07	100.0	log-count proxy of literature support	
P09	100.0	100 - sum(metabolite organ accumulation risks)	

Weak Spots (Below 60)

Principle	Score	Method	Reference
P59	30.0	Stimuli-responsive bonus by release kinetics	
P34	20.0	DNA-based carriers score high	
P43	20.0	FUS-response by carrier acoustic properties	

14e-ii. Complete 62-Principle Scoreboard (P01 → P62)

All 62 principles in canonical order, covering [Class A \(Surrogate\)](#), [Class B \(Deep Physics\)](#), and [Class C \(Translational\)](#).

P#	Cls	Title	Score	Conf.	Method
P01	B	Adversarial Stress-Testing Engine	62.0	MODERATE	Mean of 6 stress scenarios (pH4.5, pH2, pH8.5, 42°)
P02	B	Cross-Species PK Scaling (Allometric)	100.0	MODERATE	$BW^{0.75}$ scaling + class-specific adjustment
P03	A	Competitive DDS Landscape Radar	83.0	LOW	Novelty = $100 - \text{frequency_of_combo_in_CNS_trials}$,
P04	B	Quantum Coherence Transport Model	16.1	MODERATE	WKB tunneling: $P \propto \exp(-2 \cdot \text{barrier})$
P05	A	In-Silico Patient Subgroup Stratifier	70.0	MODERATE	% subgroups responding (lower CYP risk = more univ)
P06	A	Lysosomal Trafficking Predictor	72.5	MODERATE	Carrier endosomal_escape $\oplus$ drug-passive permeation
P07	A	Real-Time Literature Mining (PubMed)	100.0	LOW	log-count proxy of literature support
P08	B	Degradation Kinetics Under Oxidative S	58.3	MODERATE	Arrhenius $k=A \cdot \exp(-E_a/RT)$ for carrier; minus penal
P09	A	Digital Pharmacovigilance Engine	100.0	MODERATE	$100 - \text{sum}(\text{metabolite organ accumulation risks})$
P10	B	LNP Ionization State Predictor	60.0	LOW	Non-LNP carrier — falls back to drug-side ionizati
P11	B	Formulation Instability Fingerprint	65.0	MODERATE	$100 - \text{sum}(\text{weak-bond penalties from SMILES})$
P12	B	CNS Disease-Stage-Aware Dosing	91.5	MODERATE	Score = $100 - \text{BBB_integrity_factor} \times (0.4 + 0.6 \cdot \text{CNS-})$
P13	B	PBPK Digital Twin	100.0	MODERATE	AUC_brain/AUC_plasma proxy; 3-compartment surrogat
P14	A	In-silico Dissolution & Release Profil	98.6	MODERATE	t50 from carrier kinetics $\times$ drug-membrane partitio
P15	A	Shelf-life & Degradation Predictor	85.1	MODERATE	Arrhenius baseline $\times (0.7 + 0.3 \cdot \text{EE}) \times \text{drug_hydroly}$
P16	B	Scale-up & Manufacturability	50.0	MODERATE	Scale-readiness $\times$ shear-robustness, minus drug-pro
P17	A	Nanotoxicity & Immunogenicity Screenin	86.2	MODERATE	Mean of (hemolysis penalized by drug net+ charge,
P18	B	Active Targeting & Receptor Binding	88.0	HIGH	Ligand-affinity table $\times$ density $\times$ drug-compatibili
P19	A	QbD - Quality by Design Engine	100.0	MODERATE	Fraction of CQAs (size, zeta, EE, PDI, drug-loadin
P20	A	Cost-Efficiency Engine	61.6	LOW	Score = $100 - 3 \cdot (\$/\text{mg estimate, drug+carrier+ligan})$
P21	C	Pre-IND Regulatory Reports	0.0	—	skipped_deep_validation_insufficient
P22	A	Protein Corona Predictor	64.2	MODERATE	$100 \times \exp(-\text{corona_thickness}/10)$; thickness from ca
P23	B	Dynamic Crystal Polymorphism	88.0	MODERATE	$100 - \text{polymorph risk} (\text{rot_bonds} + \text{H-bond pattern})$
P24	B	Shear-Stress & Scale-Up Collapse	90.0	MODERATE	$\log_{10}(\text{crit_shear}/\text{operating_shear}) \times 30$; crit_shear
P25	A	Extractables & Leachables	100.0	MODERATE	$100 - \text{leachables risk by drug LogP} \times \text{carrier}$

P#	Cls	Title	Score	Conf.	Method
P26	A	Microbiome-Excipient Interactions	82.0	MODERATE	$100 - \text{mean_microbiome_degradability}(\text{carrier}) - \text{dru}$
P27	A	Lyophilization Cycle Optimizer	77.5	MODERATE	$100 - (\text{Tg} - 5) \times 1.5$
P28	A	3D-Printed Polypill Rheology	80.0	MODERATE	Carrier-specific printability index
P29	B	Biomimetic & Exosome Engineering	55.0	MODERATE	Stealth-from-macrophage score; PEG $\geq 5\%$ boost
P30	B	QM/MM Stimuli-Responsive Cleavage	45.0	MODERATE	$\text{Score} = 50 \times 7.4 - \text{pH_trigger} $
P31	B	In-Silico Biodistribution	100.0	MODERATE	$\text{brain\%} = \text{BBB} \cdot \text{ligand} \cdot \text{size_match} \cdot 100$; 100 if $\geq$
P32	C	Automated FTO & IP Evader	0.0	—	skipped_deep_validation_insufficient
P33	B	BBB Quantum Breaker (Trojan-Horse Desi	56.4	MODERATE	$(0.6 \cdot \text{ligand} + 0.4 \cdot \text{size}) \times \text{density_factor}$
P34	A	DNA Logic Gates & Bio-computing	20.0	MODERATE	DNA-based carriers score high
P35	A	Microgravity Formulation Engine	100.0	LOW	Sedimentation-Péclet proxy; smaller = better
P36	A	Geopolitical Supply-Chain Resilience	60.0	MODERATE	Supplier-diversity index by carrier + ligand
P37	A	Eco-Destructible Pharma	95.0	MODERATE	Biodegradability index by carrier class
P38	B	Glymphatic Clearance Trap	100.0	MODERATE	Stokes-Einstein: optimum 80-150 nm
P39	A	Microglial Activation & Neuroinflammat	100.0	MODERATE	$100 - (\text{cationic_charge} + \text{carrier_risk} - \text{PEG_protec})$
P40	B	Intranasal-to-Brain Delivery	65.0	MODERATE	Mucoadhesion + thermo-responsive boost
P41	B	Exosome Cargo Loading Thermodynamics	40.0	LOW	Not an exosome carrier — N/A
P42	A	Region-Specific Spatiotemporal Navigat	85.0	MODERATE	Ligand-region specificity table
P43	B	FUS-Responsive Nanocarriers	20.0	MODERATE	FUS-response by carrier acoustic properties
P44	B	CNS-Specific PBPK Time-Machine	100.0	MODERATE	AUC over 24h therapeutic window proxy
P45	C	FDA 21 CFR Part 11 Compliance	0.0	—	skipped_deep_validation_insufficient
P46	A	Polypharmacy & DDI Simulator	100.0	MODERATE	CYP-inhibition heuristic from SMILES
P47	B	Free Energy Perturbation (FEP+)	90.0	LOW	Vina-like ΔG proxy from MW + LogP (deep mode runs)
P48	A	Off-Target Toxicity & QSAR (50-recepto	100.0	MODERATE	50-receptor QSAR surrogate (hERG, 5HT2B, AhR risks)
P49	A	Organ-on-a-Chip Simulator	80.0	MODERATE	Microfluidic compatibility (size + PDI)
P50	B	Cryo-Chain Thermal Excursion Predictor	71.5	MODERATE	Distance from -20°C phase transition $\times 1.3$
P51	B	Terminal Sterilization Survivability	85.0	MODERATE	Carrier gamma-radiation survival (25 kGy)

P#	Cls	Title	Score	Conf.	Method
P52	A	Continuous Manufacturing Digital Twin	80.0	MODERATE	Continuous-process readiness by carrier
P53	A	Dark Data & Negative Results Vault	100.0	MODERATE	Similarity-to-documented-failure
P54	A	Pharmacogenomic-Guided Targeting	70.0	MODERATE	Class-based pharmacogenomic applicability
P55	C	Automated Grant & NIH Proposal Generat	0.0	—	skipped_deep_validation_insufficient
P56	C	Patentability Score Engine	0.0	—	skipped_deep_validation_insufficient
P57	B	Microfluidics & LNP Synthesis Digital	90.0	MODERATE	Microfluidic readiness for size 50-150 nm
P58	B	Impurity Cascade Predictor	85.0	MODERATE	Carrier residual-metals impurity profile
P59	B	4D Shape-Shifting Carriers	30.0	MODERATE	Stimuli-responsive bonus by release kinetics
P60	A	Swarm Nanorobotics Intelligence	15.0	LOW	Swarm capability (most carriers: no)
P61	B	Synthetic Clinical Trials & Virtual Hu	95.0	MODERATE	Virtual-cohort responder fraction
P62	A	Biobetter / Supergeneric Generator	50.0	LOW	Novelty distance from on-market CNS DDS

62 principles evaluated. Composite: 83.0/100 (EXCELLENT)

14f. Class B Deep Physics Validation (Top-1 DDS)

Verdict: MARGINAL **Pass rate:** 67.9% (19/28 principles validated, threshold 70%)

Combined score (surrogate pass-through + real physics): 19/28 principles scored PASSED (67.9%). Of these, only 7/28 principles ran independent deep computation (passed 7/7 = 100.0%); the rest re-used their Class-A surrogate score pending a future full-physics HPC run. Cite independent_pct, not pct, as evidence of physics-based validation.

Per-principle deep validation

P#	Validated	Score	Value	Conf.	Method
P01	✓	62.0	61.98	MODERATE	Full MD stress simulation — surrogate value confirmed (
P02	✓	90.0	1.5	HIGH	Resolved clinical half-life (bundle tier 1); Mahmood 20
P04	✗	16.1	0.0805	LOW	Full QM tunneling (QCElemental + ASE) — surrogate value
P08	✗	58.3	58.26	LOW	Full radical-chain reaction MD — surrogate value confir
P10	✓	60.0	60.0	MODERATE	Constant-pH MD simulation — surrogate value confirmed (
P11	✓	65.0	65	MODERATE	DFT bond dissociation energies — surrogate value confir
P12	✓	91.5	91.5	MODERATE	Stage-specific PBPK with full BBB physiology — surrogat
P13	✓	100.0	12.6429	HIGH	3-compartment PBPK ODE (scipy.integrate.odeint): blood
P16	✗	50.0	50.0	LOW	CFD of 1000L bioreactor — surrogate value confirmed (fu
P18	✓	98.0	-9.8	HIGH	MM/GBSA-style ΔG estimate from validated ligand-recepto
P23	✓	88.0	88.0	MODERATE	CSP via DFT — surrogate value confirmed (full-physics H
P24	✓	90.0	89.95	MODERATE	Full CFD + MD coupling — surrogate value confirmed (ful
P29	✗	55.0	55	LOW	Full membrane-fusion MD — surrogate value confirmed (fu
P30	✗	45.0	45.0	LOW	Full QM/MM (PySCF or ORCA) — surrogate value confirmed
P31	✓	100.0	54.05	MODERATE	7-organ whole-body distribution with size + charge + li
P33	✗	56.4	56.4	LOW	Atomistic docking + SMD pulling — surrogate value confi
P38	✓	95.4	15.6	HIGH	Stokes-Einstein $D = k_B \cdot T / (6\pi\eta r)$ at 37°C, $\eta_{CSF} = 7 \times 10^{-4}$ Pa·s
P40	✓	65.0	65	MODERATE	Full nasal cavity CFD — surrogate value confirmed (full
P41	✗	40.0	40	LOW	Full membrane mechanics MD — surrogate value confirmed
P43	✗	20.0	20	LOW	Full acoustic radiation force MD — surrogate value conf
P44	✓	100.0	47.76	HIGH	4-compartment CNS-PBPK ODE with glymphatic clearance: B
P47	✓	79.2	-7.2	LOW — LI	LIE approximation (fallback — Vina unavailable)
P50	✓	71.5	71.5	MODERATE	Full coarse-grained MD lipid phase transition — surroga
P51	✓	85.0	85	MODERATE	Full radical-chain damage MD — surrogate value confirme
P57	✓	90.0	90	MODERATE	Full CFD of mixer geometry — surrogate value confirmed
P58	✓	85.0	85	MODERATE	Full QM/MM cascade simulation — surrogate value confirm
P59	✗	30.0	30	LOW	Full coarse-grained MD morphological transition — surro

P#	Validated	Score	Value	Conf.	Method
P61	✓	95.0	95	MODERATE	Full Monte-Carlo population PBPK — surrogate value conf

14g. Class C Translational Deliverables

Principle	Title	Status	Score / Outcome
P21		skipped_deep_validation_	—
P32		skipped_deep_validation_	—
P45		skipped_deep_validation_	—
P55		skipped_deep_validation_	—
P56		skipped_deep_validation_	—

Narratives

P21: Deep validation passed only 19/28 principles. Translational deliverables withheld until a higher-ranked DDS passes deep validation.

P32: Deep validation passed only 19/28 principles. Translational deliverables withheld until a higher-ranked DDS passes deep validation.

P45: Deep validation passed only 19/28 principles. Translational deliverables withheld until a higher-ranked DDS passes deep validation.

P55: Deep validation passed only 19/28 principles. Translational deliverables withheld until a higher-ranked DDS passes deep validation.

P56: Deep validation passed only 19/28 principles. Translational deliverables withheld until a higher-ranked DDS passes deep validation.

v23: structured outlines for Pre-IND, Grant, Patent will be rendered as fully formatted Word/PDF deliverables.

14h. Top-N Fallback Audit Trail

If the Top-1 DDS fails deep validation ($\geq 70\%$ threshold), the orchestrator falls back to Top-2, then Top-3. Each candidate's outcome and the explicit transition reasoning are recorded below.

Rank	DDS Name	Verdict	Pass %	Promoted?
#1	RVG29-PLGA-NP	MARGINAL	67.9%	—
#2	Tf-SLN-Stealth	MARGINAL	60.7%	—
#3	ApoE-SLN	MARGINAL	60.7%	—

#1 RVG29-PLGA-NP — MARGINAL

Failure reason: Deep validation MARGINAL: only 19/28 principles validated (67.9%, threshold 70%). Failed principles include: P04, P08, P16, P29, P30.

Transition reason: Falling back to rank #2 because this DDS did not meet the 70% deep-validation threshold.

Failed principles: P04, P08, P16, P29, P30, P33, P41, P43

#2 Tf-SLN-Stealth — MARGINAL

Failure reason: Deep validation MARGINAL: only 17/28 principles validated (60.7%, threshold 70%). Failed principles include: P01, P04, P08, P16, P24.

Transition reason: Falling back to rank #3 because this DDS did not meet the 70% deep-validation threshold.

Failed principles: P01, P04, P08, P16, P24, P30, P33, P40

#3 ApoE-SLN — MARGINAL

Failure reason: Deep validation MARGINAL: only 17/28 principles validated (60.7%, threshold 70%). Failed principles include: P01, P04, P08, P16, P24.

Transition reason: No more candidates in Top-3 — reverting to rank #1 and reporting with MARGINAL verdict.

Failed principles: P01, P04, P08, P16, P24, P30, P33, P40

15. Executive Decision Framework

Criterion	Status	Evidence
DLVO colloidal stability	FAIL ✗	V_total = 6 kT
BBB penetration >10%	PASS ✓	DDS BBB = 100.0%
Cardiac off-target	FLAG ■	hERG/Nav/Cav QSAR
Clinical trial GO	NO-GO / REFORMULATE	Response 100% AE 91.2%
Sterilization feasible	4 method(s)	Autoclave 121C
Supply chain	HIGH	Score 0/100
OVERALL DECISION	CONDITIONAL GO	Reformulate/re-evaluate before IND-enabling studies — failed: synthetic trial returned 'NO-GO / REFORMULATE'; colloidal instability (DLVO); cardiac off-target flag

NOTE: All results are computational predictions. Required wet-lab validation: DLS/Zeta (physical stability), HPLC/NMR (drug content), BBB TEER assay, in vivo PK in rodent model.